



Attention-assisted deep neural network for nonlinear equalization and denoising in time-stretched photonic ADC

MEDHA BHAVANI GIRISH,[†] MANDEEP SINGH,[†] DEEPA VENKITESH,^{ID}
AND LAKSHMI NARASIMHAN THEAGARAJAN^{*}

Department of Electrical Engineering, Indian Institute of Technology Madras, Chennai, 600036, India

[†] Authors contributed equally

^{*} lnr@ee.iitm.ac.in

Abstract: We present a photonic time-stretch (PTS) front-end architecture whose signal recovery methodology is agnostic to both sampling rate and input signal power, thereby eliminating the need for repeated model retraining. This is implemented using a deep neural architecture that seamlessly integrates an attention-enhanced long short-term memory (LSTM) recurrent neural network and a feedforward neural network (FFNN) for suppression of noise and nonlinear distortions induced by the front-end. The two neural network models function in parallel: the attention-assisted LSTM predicts the normalized signal waveform, while the FFNN estimates the root-mean-square (RMS) signal power. The recovered waveform and power are combined for accurate reconstruction of the original signal. Unlike the previous photonic analog-to-digital converter (ADC) system implementations, the proposed LSTM-driven framework is capable of processing input signals across arbitrary sampling rates and sample sizes without the need for model retraining. Further, the attention mechanism enables effective modeling of long-term temporal dependencies by prioritizing the more relevant time steps, ensuring accurate signal recovery even for longer input sequences without introducing any signal attenuation. The proposed system achieves a signal-to-noise-and-distortion ratio (SNDR) of 63 dB and an effective number of bits (ENOB) of 10.18 using a 1 GSa/s electronic ADC for input RF frequencies in the range of 0.6 GHz to 1.4 GHz. The proposed PTS front-end reduces the ADC bandwidth requirement by threefold. This approach enables high-frequency signal digitization using low-bandwidth ADCs, resulting in improved SNDR and ENOB.

© 2026 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

1. Introduction

The demand for high-performance and high-resolution analog-to-digital converters (ADCs) with elevated sampling rates is essential for modern signal processing applications, including ultra-wideband radars, high-speed oscilloscopes, radio astronomy, and ultra-wideband signal measurements. A recent survey on photonic ADCs has shown that the theoretical limit of ENOB is around 4.5 when an electronic clock, with a timing jitter of 67 fs, samples a signal of 100 GHz bandwidth using an electronic ADC [1]. Thus, the advancement of electronic ADCs has been constrained by the intrinsic timing jitter of the sampling clock and bandwidth limitations, which ultimately limit the signal-to-noise-and-distortion ratio (SNDR) and the effective number of bits (ENOB). To address these issues, photonics-assisted time stretched (PTS) front-end that leverages the ultra-short pulse widths and spectrally rich mode-locked lasers and the wide bandwidth provided by optical modulators, have emerged as a promising solution [2]. This enables accurate digitization of high-frequency and wideband signals with improved SNDR and ENOB.

For high-bandwidth applications such as ultra-wideband (UWB) communications, continuous-time RF signal processing is required. A time-wavelength mapping architecture is used where the input RF signal modulates a spectrally broadband, pre-chirped optical pulse. The modulated signal

is further time-stretched by passing through a dispersive medium to expand the pulse duration (time-stretch). This effectively compresses the RF signal bandwidth, bringing it within the sampling capabilities of electronic digitizers. To enable continuous operation, the stretched and temporally overlapping pulses are spectrally partitioned using a wavelength-division multiplexing (WDM) filter, to allow simultaneous detection and digitization by a backend array of photodiodes and electronic ADCs (eADCs) [2]. Despite these advantages, the architecture is limited by system-induced nonlinear distortions that hinder its practical applicability. These nonlinearities primarily originate from: (i) the non-ideal response of the demultiplexing filters, which leads to imperfect channel isolation and spectral overlap; (ii) the intrinsic nonlinearity of the electro-optic modulator—although quadrature biasing suppresses even-order distortions, dispersion-induced sideband walk-off combined with square-law photodetection reintroduces these distortions; (iii) photodetector saturation and (iv) RF modulation on a Gaussian-shaped optical pulse, which necessitates envelope correction for accurate reconstruction of the original RF waveform. RF power fading and modulation nonlinearities have been mitigated by employing single-sideband modulation with dual-output modulators [2,3], differential operation on dual-output [4,5], asymmetric dual-parallel MZMs [6], and envelope correction techniques. Additionally, a redundancy detection method proposed in [7] addresses the distortions caused by the filter roll-off, which introduces cross-talk among the parallel channels.

1.1. Comparison of PADC architectures

An exhaustive summary of PADC implementations and the achieved specifications are given in the [Supplement 1](#). Table 1 summarizes prior work in terms of input frequency, sampling rate, and ENOB, including recent machine-learning-assisted approaches.

Table 1. Performance Comparison with State-of-the-Art Photonic ADCs

Reference	Year	Input Freq. (GHz)	Rate (GS/s)	ENOB (bits)	System Type
[8]	2019	up to 23.3	20	4.66/7.28 @ 3.44 GHz	Photonic sampled ADC with DL correction
				4.53/7.07 @ 21.13 GHz	
				4.57/9.24 @ 23.3 GHz	
[9]	2025	6	12	6.1; (7.13, 7.3) ML @ 1.4 GHz	Photonic sampled ADC + ML
Our work	2026	up to 1.4	1	10.18	Photonic time-stretched ADC + ML

As shown in Table 1, earlier ML-assisted photonic sampled ADCs such as [8] achieve operation over a wide bandwidth (up to 23.3 GHz) with high sampling rates (20 GS/s), but exhibit relatively low baseline ENOB (~ 4.5 bits), which improves to approximately 7–9 bits after deep-learning-based correction. More recent work [9] demonstrates moderate-frequency operation (up to 6 GHz) with 12 GS/s sampling, achieving ENOB values around 6.1 bits, which increase to ~ 7.3 bits with ML-based enhancement at 1.4 GHz. In comparison, the proposed continuous time-stretch photonic ADC (CTS-PADC) achieves a performance better than the state-of-the-art. Despite operating with an electronic backend of only 1 GS/s, the system attains an ENOB of 10.13 at 1.4 GHz after machine learning-based nonlinear compensation. These results highlight that, our technique has achieved an operating frequency of ≈ 2.8 times the Nyquist bandwidth of the sampler, with an ENOB of 10.18. As summarized in Table 1, the CTS-PADC demonstrates a favorable trade-off between sampling rate and ENOB, establishing competitive performance for high-resolution photonic ADC applications.

The time-stretching media in the CTS-PADC, inherently introduces chromatic dispersion that creates inter-symbol interference; hence, it acts as a system with memory. Additionally, the input signal is inherently correlated across time. Moreover, the frequency responses of the optical

filters used in the experiment overlap, thus leading to correlation between the corresponding time samples, after they are stitched (the input spectrum and the spectral response of these filters are shown in Fig. 1(b); this figure clearly indicates the overlap of the filtered spectra, leading to correlations). Thus, any recovery method or equalizer requires memory to compensate for the distortions. Further, due to the nonlinear responses of the MZM and optical fiber, the nature of these distortions are non-linear. Thus, denoising or recovering the original continuous signal from this setup is equivalent to performing time-series denoising by a technique that is sample-size-agnostic.

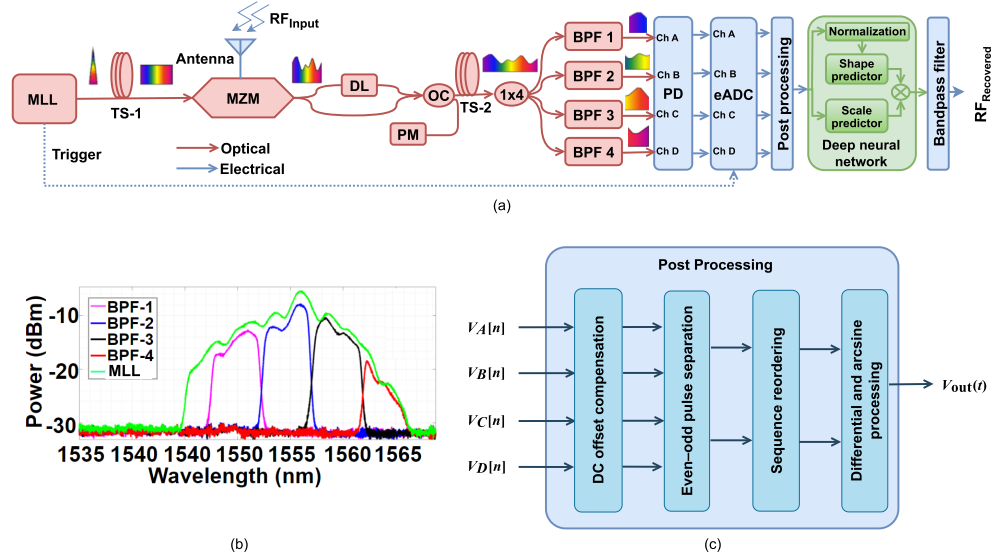


Fig. 1. (a) Schematic of the proposed PTS frontend, (b) measured spectra of the MLL and WDM filters, and (c) corresponding post-processing steps.

For the choice of neural architecture to use, Table 2 summarizes the different possibilities of models to select from and compares them in terms of size, adaptability, and memory. Among the applicable neural architectures, we have chosen the one that provides a suitable balance between efficiency and model complexity.

Table 2. Comparison of different neural network architectures

Architecture	No. of Parameters	Variable Input Length	Temporal Context Size
MLP/FFNN	High	No	N/A
1-D CNN	Low	No	N/A
GRU	Low	Yes	Small
GRU + Attention	High	Yes	Large
LSTM	Low	Yes	Large
LSTM + Attention	High	Yes	Very Large
Transformers & its variants	Very High	Yes (by tokenization)	Very Large
Spiking Neural Network	High	No	N/A
Diffusion Models	Very High	No	N/A

The goal of this work is to design a sample-size-agnostic neural network to equalize front-end distortions in a CTS-PADC system. Since the input signal is streaming, we may not know apriori

the input sample length or sampling frequency. Further, we need a neural network with a larger context or temporal memory to exploit the temporal correlations in the input signal to denoise or equalize severely corrupted samples.

Multi-layer perceptron (MLP) or feed forward neural networks (FFNN) are not suitable for arbitrary sequence lengths of the input RF signal, because it processes input data by taking in a block of fixed number of samples at a time and treats each block of input samples independently, and lacks temporal context. Therefore, even though their denoising performance may be good, [10], when there is a change in the sampling rate or the number of samples per block, FFNN requires retraining of the model. The same is true for spiking neural networks [11], and more recent stochastic models such as diffusion models [12]. Even though convolutional neural networks (CNN) such as 1D-CNN, require fewer number of parameters, they suffer from the same problem as above. Thus, all these neural network architectures are not sample-size-agnostic and are not suitable for streaming input data.

On the other hand, recurrent neural network (RNN) architectures such as gated recurrent units (GRU) and LSTMs are suitable for streaming, sample-size-agnostic applications. Furthermore, it is known that LSTMs offer a clear advantage over GRUs [13] in terms of their capacity to model complex time-series and tracking over relatively larger temporal context lengths. However, we observed that the temporal context sizes offered by LSTMs were not sufficient for CTS-PADC equalization (as evidenced later in Fig. 6(b) and explained in detail in Section 4). Therefore, we proposed to add an attention layer to improve its temporal context length; it is known that attention [14] learns only the relevant features of temporal data, thereby retaining only useful temporal data in the LSTM cells and increasing the processable context length. But this performance comes at the cost of computational complexity, the number of parameters in our LSTM network is approximately 3.2×10^5 and the number of parameters in one multi-head attention layer alone is approximately 3.2×10^6 , which is ten times more than LSTM.

Modern transformers and their variants [15] are known to be excellent in time-series prediction tasks. They can be made sample-size-agnostic by breaking the input signal into small blocks or tokens for processing. However, the main disadvantage of any variant of the transformer architecture is the computational efficiency. Since all transformer architectures use multiple attention layers along with an FFNN at the end, the number of parameters that need to be trained (and consequently, the amount of training data required) are orders of magnitude higher than the proposed architecture.

Additionally, the PTS frontend is expected to operate with a large dynamic range. However, the normalization routines commonly used in neural networks cause poor recovery of signal magnitude. Thus, conventional neural architectures could reconstruct the shape of the input signals very well but not their magnitudes.

To overcome these limitations, we need special architectures that can cater to waveform reconstruction and power estimation with arbitrary sampling rates and sequence lengths. The proposed LSTM-attention model provides the right balance between equalization performance and computational complexity among all current neural architectures.

This work proposes a novel architecture by integrating two models in parallel - an LSTM with attention shape predictor, and an FFNN-based magnitude predictor - to independently predict the normalized waveform and the power of the signal, and combine these two to get the recovered signal. Using an LSTM additionally allows variable lengths of input data, giving us the advantage of continuous data processing. The trained model has been tested on unseen frequencies in the range: 0.6 to 1.4 GHz. The proposed model provides a good recovery performance, showing a maximum frequency recovery mismatch of 1.9 MHz and a maximum power recovery mismatch of 2.5 dBm, indicating a good generalization capability for the network. Additionally, the model is expected to maintain the same performance across higher sampling rates and bandwidths.

Since it operates directly on discrete samples rather than the underlying physical time scale, the model's predictions are independent of bandwidth and sampling rates.

The remainder of this manuscript is organized as follows. Section 2 describes the complete system design, including the experimental setup and the details of the proposed DNN for mitigating distortions introduced in the photonic frontend. Section 3 presents the experimental results and corresponding discussions. Finally, Section 4 concludes the paper.

2. System design

2.1. Photonic front-end

The block diagram of the proposed PTS frontend system is shown in Fig. 1(a). The photonic front-end has a mode-locked laser (MLL) that generates a Gaussian pulse train having a 50 MHz repetition rate, 200 fs pulse width, 20 nm spectral width, and center wavelength around 1556 nm. The output of the MLL is then passed through a time-stretching (TS-1) medium with a net dispersion of 594.37 ps/nm, resulting in an aperture of approximately 20 ns, thus ensuring non-overlapping pulses at this stage, and thereby enabling continuous-time operation. A dual-output MZM modulates high-frequency RF signals onto the stretched pulses, producing two complementary outputs. One output passes through a delay line (DL) to create a temporal shift of $T_{\text{rep}}/2$. These complementary signals are then combined in a 50:50 optical coupler to achieve time-interleaving; the reference complementary signal further facilitates envelope correction in the post-processing. The time-interleaved pulses further undergo another time stretching (TS-2) with a fiber of net dispersion of 1222.51 ps/nm. The overall stretch factor (S) of the system is approximately 3.132. The time-stretched optical signal is divided into four channels using a 1×4 splitter and subsequently filtered by four bandpass filters (BPFs) centered at 1549.71 nm, 1554.53 nm, 1559.38 nm, and 1564.27 nm, respectively, each with a 5 nm spectral bandwidth. The output spectrum of the mode-locked laser and those from the bandpass filters are shown in Fig. 1(b). A four-channel photodetector (PD) with 500 MHz electrical bandwidth converts the time-stretched optical waveforms to four parallel baseband electrical signals which are then sampled by the four channel ADC operating at a sampling rate of 1 GSa/s.

2.2. Post processing

To reconstruct the waveform, digital post-processing pipeline is proposed and it is summarized in Fig. 1(c). The static offsets introduced by the optoelectronic front-end are first compensated by applying an appropriate DC shift to each channel, yielding the discrete-time input sequences $V_A[n]$, $V_B[n]$, $V_C[n]$, $V_D[n]$ corresponding to channels A–D, respectively. The samples from the two complementary ports of the Mach–Zehnder modulator are then digitally de-interleaved and stitched for each channel to reconstruct a single high-resolution waveform. For each channel $k \in \{A, B, C, D\}$, an “even–odd pulse separation” block generates two subsequences by decimating the stitched stream:

$$V_{\text{even},k}[n] = V_k[2n], \quad (1)$$

$$V_{\text{odd},k}[n] = V_k[2n + 1]. \quad (2)$$

These subsequences represent, for each channel, the even- and odd-indexed pulses originating from the two complementary optical outputs of the modulator. The even subsequences are then combined by a “sequence reordering” block to form a single composite even sequence

$$V_{\text{even}}(t) = [V_{\text{even},A}(t), V_{\text{even},B}(t), V_{\text{even},C}(t), V_{\text{even},D}(t)], \quad (3)$$

and, analogously, the odd subsequences are stacked to form

$$V_{\text{odd}}(t) = [V_{\text{odd},A}(t), V_{\text{odd},B}(t), V_{\text{odd},C}(t), V_{\text{odd},D}(t)]. \quad (4)$$

This step arranges the per-channel information into two global streams that represent the even and odd pulse sets of the overall time-stretched waveform. To linearize the modulator response and suppress common-mode distortions, a “differential and arcsine processing” block operates on the composite even and odd sequences [4]. A differential signal is first formed as

$$V_{\text{diff}}(t) = \frac{V_{\text{odd}}(t) - V_{\text{even}}(t)}{V_{\text{odd}}(t) + V_{\text{even}}(t)}, \quad (5)$$

which effectively removes common-mode components and linearizes the interferometric transfer function produced by the complementary modulator ports. An arcsine operation is then applied to recover a waveform proportional to the original RF signal [4]:

$$V_{\text{out}}(t) = \sin^{-1}(V_{\text{diff}}(t)). \quad (6)$$

Together, the differential operation in (5) and the arcsine operation in (6) are carried out to invert modulator’s transfer function with Gaussian pulse shape.

2.3. Deep neural network (DNN)

The downsampled waveform $V_{\text{out}}(t)$ is further refined using a data-driven equalization stage. Specifically, the signal is processed by a deep neural network to compensate for residual nonlinearities arising from imperfect demultiplexing in WDM filters, modulator-induced distortions, photodetector saturation, and RF modulation of Gaussian-shaped optical pulses, which require envelope correction for accurate signal recovery. This learning-based post-processing mitigates nonlinear distortions in the reconstructed RF signal obtained from low-bandwidth ADC samples. There are four main challenges in designing a neural network for reconstructing the original signal from this distorted signal: (1) the neural network should be capable of supporting multiple sampling rates without the need for retraining, (2) in order to enable denoising, the neural network should be able to identify the appropriate temporal features of the original signal and distinguish them from noise or distortion, (3) the neural network should be capable of processing data whose magnitude change by orders of magnitude, i.e., having high dynamic range, and (4) the ability to remove out-of-band noise.

We propose to address the above issues as follows: (1) We design an long short-term memory (LSTM) based deep recurrent neural network (DRNN) that supports input at arbitrary sampling rates and lengths [16] without requiring any retraining. (2) We employ multi-head attention models in our neural architecture to enable the neural network to dynamically focus on the signal features in the input data. (3) To handle inputs of different magnitude, we propose a novel deep neural architecture which decouples time-domain signal reconstruction and power estimation into two parallel tasks - we refer to this as shape prediction and scale prediction tasks. The shape prediction is performed by the LSTM, while the scale prediction is performed by an FFNN. (4) After the neural network based reconstruction, the out-of-band noise is reduced using a reconfigurable bandpass filter. This architecture is illustrated in Fig. 2 and the implementation details of it are given below.

2.3.1. Shape predictor

The shape predictor compensates for time-domain waveform distortions introduced by the photonic front end. The input of the shape predictor is magnitude-normalized signal. Thus, its output does not contain any magnitude or power information of the original signal, but only the signal’s shape information is recovered. The shape predictor’s architecture is described below. *Long Short-Term Memory (LSTM) Networks*: The shape predictor is realized as an LSTM network augmented with attention layers. LSTMs are a specialized type of deep recurrent neural network (RNN) designed to address the long-term and short-term temporal dependencies between

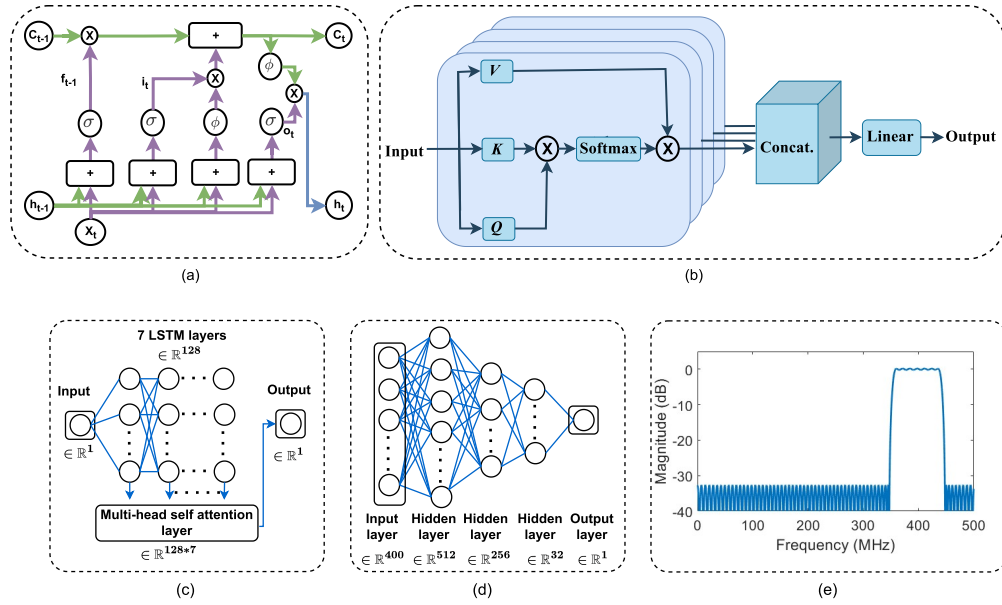


Fig. 2. Neural network building blocks: (a) LSTM cell, (b) multi-head self-attention, (c) shape predictor, (d) scale predictor, and (e) illustrative frequency response of the reconfigurable bandpass filter.

different time steps of a signal that varies over time [16]. Further, LSTMs can produce output sequences of any length; this overcomes the limitations of FFNNs that produces only fixed length output sequence of fixed sampling rate.

The LSTM network consists of cells, which contain a set of gates that regulate the flow of information into and out of the cell. The gates within an LSTM cell learn to retain relevant temporal correlation patterns while discarding irrelevant details, allowing the LSTM cell to store temporal features over extended periods. It has three different kinds of gates: an input gate (i_t), an output gate (o_t), and a forget gate (f_t). The long-term memory to aggregate related data across time steps is captured by the cell state vector (C_t) [16]. The structure of the single LSTM cell is shown in Fig. 2(a). The samples from the photonic front-end is fed as the input X_t to the LSTM, sequentially over time. The output of a single LSTM cell is computed as follows:

$$\begin{aligned}
 i_t &= \sigma(W^i x_t + U^i h_{t-1} + b^i), \\
 f_t &= \sigma(W^f x_t + U^f h_{t-1} + b^f), \\
 o_t &= \sigma(W^o x_t + U^o h_{t-1} + b^o), \\
 C_t &= f_t \odot C_{t-1} + i_t \odot \phi(W^c x_t + U^c h_{t-1} + b^c), \\
 h_t &= o_t \odot \phi(C_t),
 \end{aligned}$$

where $\odot$ is element wise multiplication, ϕ and σ are the tanh and sigmoid activation functions, W, U are weight matrices, b are the bias terms, t and $t-1$ indicate the current and previous time steps, respectively. Here, the input is $x_t \in \mathbb{R}$, the activation vectors are $f_t, i_t, o_t \in (0, 1)^{n_h}$, the state vector is $C_t \in \mathbb{R}^{n_h}$, the output vector is h_t , and n_h is the number of LSTM cells. The LSTM model processes the input $x_t \in \mathbb{R}$ at time step t to generate an output h_t at time step t . Thus, as LSTMs accept only one signal sample at a time, non-linearly processes them along with the samples in

memory and produces one sample output at a time, they can handle arbitrary sampling rates and sequence lengths without retraining.

Attention mechanism: The attention mechanism enhances sequence modeling by enabling the network to explicitly learn which time steps in an input sequence are most relevant for predicting each output, rather than compressing all temporal information into a single hidden representation. This helps us to reconstruct signals even when samples at certain time instances are severely distorted and, therefore, retain the correct neural network context for future signal reconstruction. The internal structure of a multi-head self-attention layer is shown in Fig. 2(b). The input sequence is linearly projected into query (Q), key (K), and value (V) representations [17]. Similarity scores are computed via a dot product ($\times$) between Q and K, followed by a Softmax operation that normalizes these scores into attention weights. These weights are applied to the value vectors through a second multiplication ($\times$), producing a weighted sum that emphasizes relevant input features.

The stacked blocks in the diagram denote multiple attention heads operating in parallel, each learning independent representations. The outputs from all heads are concatenated (Concat) and passed through a final linear layer (Linear) to yield a context-rich output representation. In the proposed architecture, the attention layer operates on the concatenated outputs of multiple LSTM layers, enabling the model to recover and exploit temporal dependencies that may otherwise be attenuated, particularly for long sequences. This is illustrated in Fig. 2(c). The LSTM with attention based shape predictor takes the normalized distorted signal coming from the photonic front-end, and outputs a normalized signal with the distortions equalized.

2.3.2. Scale predictor

The scale predictor outputs a single scalar that reflects the signal's absolute power. Since normalization removes the magnitude information, the scale predictor operates on unnormalized inputs to estimate the true signal power. At inference time, the predicted RMS value is used to rescale the normalized waveform output, thereby recovering both the correct magnitude and the temporal waveform of the RF signal.

The scale predictor is modeled as an FFNN that predicts a single scalar value, i.e., the power, from the input signal, as shown in Fig. 2(d). FFNNs are a class of artificial neural networks in which information propagates uni-directionally from the input layer to the output layer through hidden layers [16]. Each neuron computes a weighted sum of its inputs followed by a nonlinear activation function. The parameters of the FFNN are optimized during training using backpropagation to minimize the loss function. We use FFNN to predict the power of the recovered signal from the nonlinearly distorted signal, and this scalar value is used to fix the power of the signal predicted by the LSTM. This takes a section of the distorted signal as input and predicts a single scalar power, trained using the true powers of the input to the photonic front-end as the label. Thus, the FFNN serves as a non-linear regression model for the signal's power prediction.

Despite these shape and scale corrections, the final recovered signal remains limited by the inherent system noise, which ultimately degrades the SNDR and ENOB. To minimize the effect of noise, we present a reconfigurable bandpass filter; this is discussed next.

2.4. Reconfigurable bandpass filter

Often, the recovered signal's bandwidth is less than the total bandwidth of the ADC. This results in out-of-band noise causing temporal distortions. However, the center frequency of the signal is unknown during reconstruction. To suppress the out-of-band noise introduced by the photonic front-end and improve the SNDR of the recovered signal, we implement a digital reconfigurable bandpass filter. We use a Finite Impulse Response (FIR) filter designed through the Parks–McClellan method [18, p. 498] that gives a linear phase filter. The filter is configured

with a bandwidth of 80 MHz and a transition width of 10 MHz on each side, employing 302 taps at a sampling rate of 2 GSa/s. The choice of the filter bandwidth is based on the maximum bandwidth of the input signal. When the maximum bandwidth of the input signals changes, the filter taps need to be optimized accordingly along with the proposed architecture. Whereas, when the sampling rate changes only interpolation or decimation of the filter taps is required. The center frequency of the filter is dynamically and periodically reconfigured by identifying the dominant frequency component of the recovered signal through a fixed length fast Fourier transform of the LSTM's output. The shifting of the center frequency does not require a complete recalculation of the filter coefficients; multiplying the filter tap coefficients by a sinusoid of frequency equal to the shift amount will result in the required BPF.

The frequency response of the digital reconfigurable bandpass filter, shown in Fig. 2(e), demonstrates an optimal equiripple characteristic achieved using the Parks–McClellan (Remez exchange) algorithm. Centered at the signal's dominant frequency, the response exhibits a flat 80 MHz passband in which the approximation error is uniformly distributed, resulting in small, consistent ripples rather than the decaying oscillations typically observed in window-based filter designs. The stopbands show deep and uniform attenuation of about 32 dB, which is critical for suppressing the wideband noise floor inherent to the photonic frontend. By minimizing the maximum deviation from the ideal brick-wall response, the filter maximizes the SNDR of the recovered waveform while preserving the phase linearity required for accurate time-stretched signal reconstruction. Note that the BPF operates in the time-stretched frequency scale.

Thus, the overall proposed system first processes the input RF signal through a photonic time-stretch front end to reduce the effective bandwidth, then employs an attention-enhanced LSTM with a parallel RMS scale predictor to compensate for nonlinear distortions, followed by reconfigurable bandpass filtering to improve the SNDR and ENOB. The recovered RF frequency is related to the original input frequency by the time-stretch factor (S) as: $RF_{\text{recovered}} = RF_{\text{input}}/S$, where the time-stretch operation scales the temporal features and compresses the spectral content by the same factor, enabling the use of low-rate electronic ADCs for instantaneous measurement of ultra-wideband RF signals.

3. Training the shape and scale predictors

The LSTM neural network with attention and the FFNN were trained using the time-series data experimentally collected from the photonic frontend. We collected data for RF frequencies in the range 0.6 to 1.4 GHz, with a uniform step of 25 MHz. For each frequency, the RF power is also varied from −14 dBm to 12 dBm in steps of 2 dBm. This time series data along with the ground truth data, which were generated numerically, are used to train the networks. The dataset is split into two parts: 80 % for training and 20 % for validation and testing. The input signal is normalized using z-score normalization with mean 0 and standard deviation 1, and then passed through the model. The LSTM with attention model is optimized using the Adam optimizer with a learning rate of 0.0008, and the loss function employed is mean squared error (MSE) which is used for predicting the real-valued RF frequencies. The MSE is calculated by taking the squared difference between each sample of the output and the target sequence and then summed over all the samples. As seen in Fig. 2(c), after tuning the hyperparameters, the trained LSTM-RNN with attention architecture is configured with the following settings: The input is of dimension 1, followed by 7 hidden LSTM layers of size 128. The number of LSTM layers were carefully chosen by picking the least number that provided the best performance. This is further followed by a single multi-head self-attention layer, having 4 heads. This attention layer acts on the concatenated output of each of the LSTM layers. This is followed by a final fully connected layer that gives an output of dimension 1. The output from the LSTM network with attention is further scaled in magnitude by the output of the FFNN. The FFNN is trained on the same dataset. It uses a sequence of length 400 samples from the input signal and outputs a scalar representing

the RMS power of the recovered signal. This FFNN model is trained using the AdamW optimizer with a learning rate of 1.19×10^{-4} and L2 weight regularization of 3.55×10^{-4} . As seen in Fig. 2(d), the FFNN architecture accepts an input signal of length 400 samples, and consists of four linear layers with dimensionalities 512, 256, 32, and 1 respectively, with each layer followed by a LeakyReLU activation function, enabling hierarchical feature extraction and dimensionality reduction.

The loss function of the trained LSTM with attention architecture is depicted in Fig. 3. The model's improved performance with additional training iterations is seen by the declining training and validation losses as the number of epochs increases. On scaling the LSTM output signal by the scalar RMS outputted by the FFNN, the recovered signal is passed through a reconfigurable bandpass filter. FFT is performed on the recovered signal to find the dominant frequency, and the reconfigurable bandpass filter is aligned accordingly to filter out the noise.

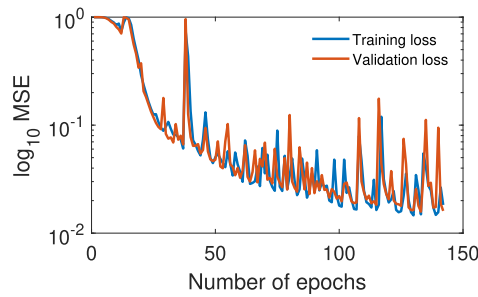


Fig. 3. Loss of the trained LSTM with attention network

4. Results and discussions

4.1. Dynamic range and linearity

As a first step toward characterizing system performance, the dynamic range and linearity of the photonic time-stretching frontend are evaluated prior to any recovery processing using the third-order intermodulation distortion (IMD3), as shown in Fig. 4(a).

To estimate the spurious-free dynamic range (SFDR) and the link gain, a two-tone test is conducted with a tone separation of 2 MHz. For this measurement, we use a similar experimental setup to the one shown in Fig. 1(a), except that the MLL is replaced with a continuous-wave (CW) laser modulated with two RF tones. We further maintain the average input optical power at values similar to that of the PTS frontend. By carrying out this test, the system's ability to handle high-power signals without introducing substantial distortions can be evaluated. We observe the noise floor and the link gain of the system to be about -110 dBm/Hz and -15.37 dB, respectively. As observed, the system exhibits linear behavior up to an input RF power of 3 dBm, beyond which nonlinear behavior is observed. The analysis considers nonlinearities introduced only by the modulator, RF amplifier, and photodetector. However, as seen in Fig. 4(b), the input RF power in the PTS frontend does not vary linearly with the output RF power of the processed signal after differential and arcsine operations. This non-linearity can be attributed to several factors including the stitching errors between adjacent channels, the arcsine operation, and modulation on a Gaussian pulse rather than CW. The system sensitivity is however primarily limited by polarization-induced amplitude jitter across the pulse spectrum, which establishes the noise floor for the minimum detectable power. In contrast, the upper bound of the dynamic range is determined by the maximum RF drive power that can be applied to the modulator. Furthermore, owing to the trained neural network-based compensation, the onset of nonlinearity does not constitute the dominant performance bottleneck.

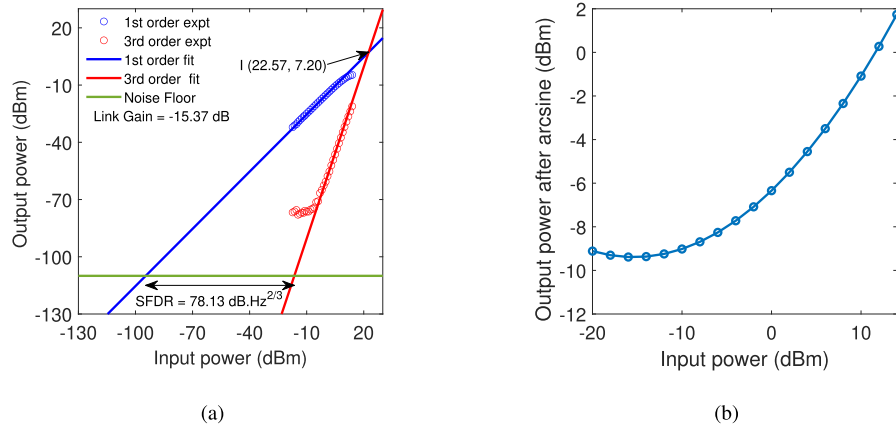


Fig. 4. Performance evaluation of the proposed PTS frontend (a) output RF power of the fundamental and third-harmonic components as a function of input RF power, measured using a CW laser with the same average power as the PTS frontend (b) output RF power of the PTS frontend after the arcsine processing plotted versus input RF power.

The proposed photonic front-end is experimentally demonstrated using input RF signals with frequencies ranging from 0.6 GHz to 1.4 GHz at different input power levels ranging from -14 dBm to 12 dBm. The samples from the PTS frontend that are experimentally collected are fed to the LSTM with attention and FFNN for the RF signal recovery. The top plot in Fig. 5(a) presents the RF spectrum obtained after time-domain expansion through the photonic frontend and post processed using an arcsine operation. Here, for illustration, an RF signal with 0.875 GHz tone at 12 dBm is fed as input. This input is spectrally scaled down by the photonic front-end by a stretch of 3.132 , resulting in an RF of 265 MHz. However, due to the nonlinearities in the photonic front-end, the spectrum at the end of the photonic front-end can be seen to contain several side lobes. This makes it difficult to distinguish the desired RF from spurious tones. Thus, the SNDR and ENOB are ultimately limited by the distortions introduced by the system.

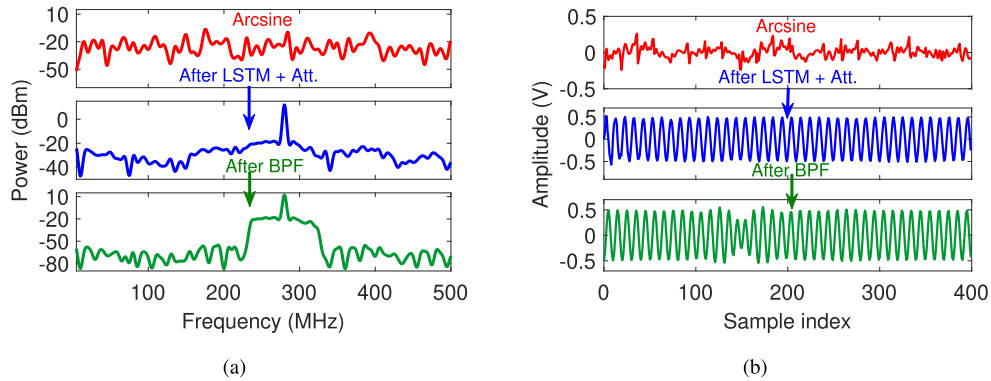


Fig. 5. (a) Spectrum of recovered RF after the arcsine operation, LSTM with attention layer and after the bandpass filter (b) Corresponding recovered signals in time domain

4.2. Validation and SNDR improvement

The performance of the trained model is demonstrated as follows. The time series data corresponding to power 12 dBm and input RF frequency 0.875 GHz is fed into the trained model

for predictions. The spectrum obtained after passing the signal through the LSTM network with attention layer is shown in blue in Fig. 5(a). It is seen that the signal recovered has a significant reduction of the spurious tones compared to the input, showing an improvement in SNDR of 32 dB. Note that SNDR is computed as : $SNDR = 10 \log_{10}(P_{Signal}/(P_{Noise} + P_{Distortions}))$, where P_{Signal} is the power of the fundamental frequency component, P_{Noise} is the integrated noise power and $P_{Distortion}$ is the total distortion power introduced by the system. The dominant frequency predicted on recovery is 279.6 MHz (scaled by the stretch factor), and the power predicted is 12.1 dBm. The green curve in Fig. 5(a) shows the spectrum of the signal on passing it through the deep neural network and further through the reconfigurable bandpass filter centered at 279.65 MHz. As observed, the noise in the side lobes are further suppressed, leading to an overall SNDR of 63 dB. From the relation $ENOB = (1.662 - SNDR)/6.02$, for SNDR of 63 dB, we get an ENOB of 10.18. It has to be noted that this SNDR of 63 dB is an improvement of 39 dB in SNDR over the previously reported photonic ADC system [10]. Figure 5(b) shows the time domain representation of these signals before recovery (red), after passing through the model (blue), and after passing through the bandpass filter (green). The pure sinusoidal signal is recovered in the domain, corresponding to a peak of 279.6 MHz in the frequency domain.

4.3. Arbitrary sequence length processing

The trained LSTM network is also tested by choosing arbitrary sequence lengths of the time domain RF frequencies. Figure 6(a) depicts that the trained model can predict the recovered frequency precisely even for different lengths of input sequences: 400, 700 and 1036. This proves the advantage of using LSTM for any arbitrary sampling rate and sequence length of the input RF compared to the state-of-art [10] because it processes each input sample sequentially and retains temporal context. The importance of the attention mechanism is illustrated through Fig. 6(b). On passing the distorted signal through the LSTM model without the final attention layer, it is observed that the LSTM output decays over time as it is unable to exploit the temporal correlations from samples that are further into the past. This attenuation was observed in majority of the sequences of duration 1000 samples or longer. Simply increasing the number of LSTM layers does not correct this, as deeper layers lead to vanishing gradients. Attention has the ability to learn only the useful features from the past input, thereby retaining only the relevant temporal data in the LSTM cells, increasing its equalization performance. As expected, on passing this same signal through the LSTM model with attention, the model is able to recover the input sinusoid with much higher accuracy, maintaining longer temporal context and preventing attenuation.

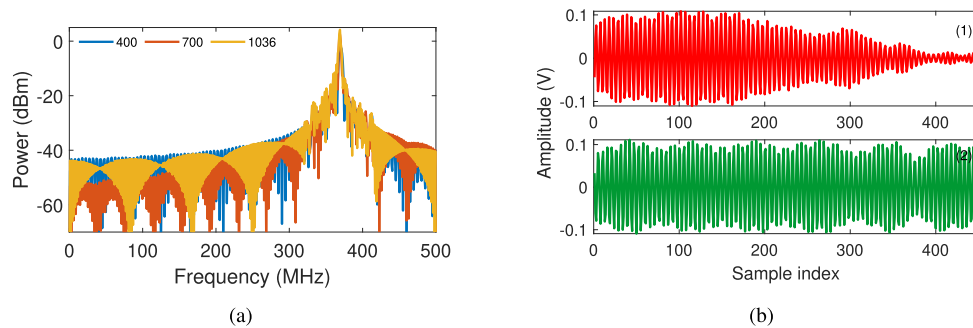


Fig. 6. (a) Recovered RF spectrum for varying sequence lengths, (b) Recovered time domain signal without attention in (1) and with attention in (2).

4.4. RF power recovery accuracy

The accuracy of the recovered RF power is also analyzed. For illustration, the power of the recovered RF versus the input RF power at input RF frequencies of 0.6 GHz, 0.9 GHz and 1.4 GHz are shown in Fig. 7(a). It is observed that the recovered RF power closely matches that of the input with an average offset of < 0.5 dBm, and a maximum offset of < 2.5 dBm. Figure 7(b) shows the error in power prediction for these 3 frequencies. The trained model matches the recovered RF power with the input RF power at a reasonable accuracy, thereby effectively accounting for all the losses introduced by the front-end.

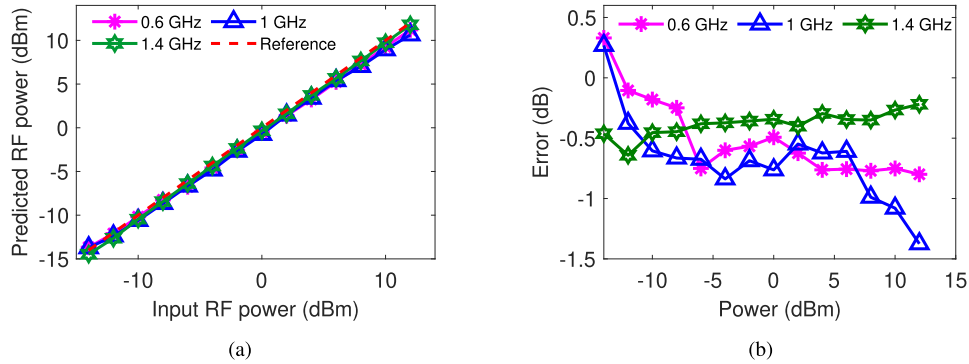


Fig. 7. (a) Accuracy of power recovery shown for different input RF frequencies, (b) Error between expected and predicted power at these frequencies.

4.5. Full band analysis

In order to analyze the performance of the system over a wide frequency range, RF input from 0.6 GHz to 1.4 GHz, at 12 dBm with a step of 25 MHz is fed to the model for recovery. The plots in the first row of Fig. 8(a) shows the spectrum of the time stretched signal with arcsine post processing. Due to the distortions introduced by the photonic frontend, it has multiple tones appearing at multiple spectral bands. Clearly, the desired RF signal (which is a single tone) is quite hard to detect in this case. It can be observed that the trained LSTM model corrects for the distortions and recovers the RF signal without many spurious side lobes; this is shown in the second row of Fig. 8(a). The average frequency offset observed in the recovered RF signal is < 0.06 MHz, and the maximum offset is < 1.9 MHz. As the frequency resolution is 5 MHz here, this frequency error is well within the acceptable limits. On applying the reconfigurable bandpass filter, the noise floor is further brought down, and the region of higher powers around the recovered RF signal frequencies represent the frequencies lying within the band of the filter; this is shown in the third row of Fig. 8(a). Figure 8(b) shows the SNDR for the signals corresponding to those in Fig. 8(a). With arcsine post processing, the average SNDR is about 3 dB, and the average SNDR after passing it through the LSTM with attention model improves to about 35 dB. There is a significant improvement on adding the reconfigurable bandpass filter, bringing the average SNDR up to about 63 dB. These SNDR values are similar across all powers and frequencies, as evident by the uniform colors in the Fig. 8.

Error Type	Maximum	Average
Frequency	1.9 MHz	0.5 MHz
Power	2.5 dBm	0.5 dBm

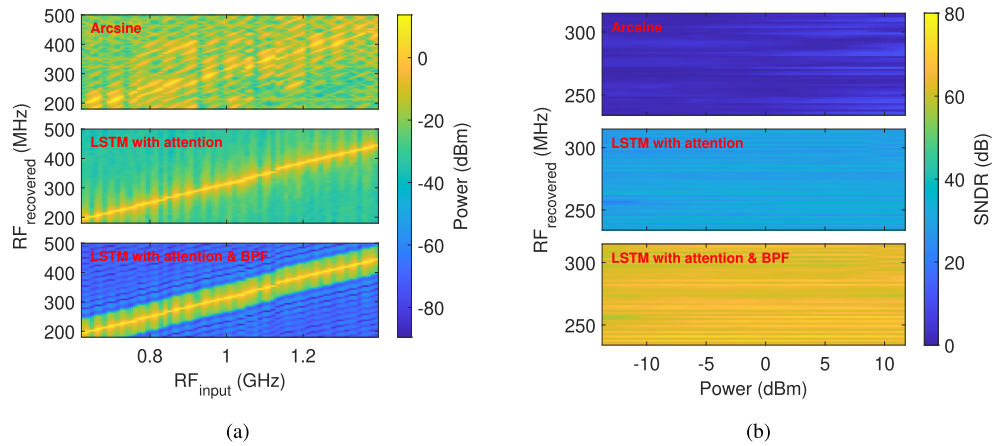


Fig. 8. (a) Parity plots for an input RF power of 14 dBm; (b) SNDR versus input power levels and recovered RF frequencies for different architectures.

4.6. Generalization capability

Further, the generalization capability of the trained model is validated by evaluating its performance on completely unseen data. During training, RF frequencies of 0.611, 0.825, 1.035, 1.203, 1.311, and 1.403 GHz, along with all their power levels from -14 dBm to $+12$ dBm, were excluded from training and validation datasets to ensure that these samples remained entirely unseen by the model. After training, the model was evaluated on this unseen dataset of about 5100 samples, and the results are presented in Fig. 9.

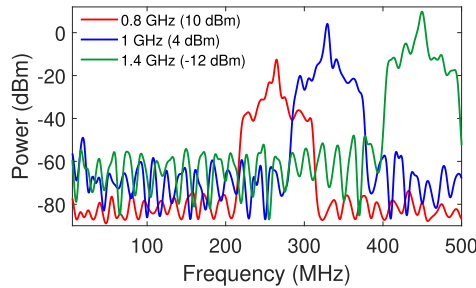


Fig. 9. Spectrum of the recovered RF for unseen data

The spectra of the recovered signals corresponding to input frequencies of 0.825 GHz (-12 dBm), 1.035 GHz (4 dBm), and 1.403 GHz (10 dBm), which were not included during the training stage are shown in Fig. 9. The recovered frequency tones after a stretch of 3.132 are observed at 264.73 MHz, 329.04 MHz, and 449.79 MHz, with frequency errors of 1.3 MHz, 0.2 MHz, and 1.8 MHz, respectively. The corresponding power deviations are 0.5 dB, 0.1 dB, and 0.2 dB, respectively. The error in both frequency and power estimation across the testing dataset are well within the acceptable limits as well, being 4.2 MHz and 1.1 dBm respectively, thereby demonstrating the strong generalization capability of the proposed model. Few-shot training can also be carried out by tuning the parameters of this model based on a small set of new data to adapt to different types of signals as well.

4.7. Latency and power consumption

Furthermore, the latency of the proposed DNN is estimated by evaluating the forward-pass execution, with the LSTM and FFNN operating independently and in parallel. The overall latency is dominated by the attention layer. The end-to-end computation requires ≈ 342.48 million multiply–accumulate operations (MACs) per 400-sample sequence, resulting in compute-only latencies of ≈ 342 ms on a 1-GMAC/s CPU, but would require only 0.27 ms on a recent model GPU [19] and 1.2 ms on a recent model FPGA [20]. Note that the latency calculated are indicative in the aforementioned compute environments, derived from the respective data sheets and not results from hardware implementations.

The overall power consumption in photonic ADC systems strongly depends on implementation-specific factors such as the mode-locked laser source, optical amplifiers, RF amplifiers, photodetectors, and backend electronic hardware, making direct comparison across different platforms challenging. It may not be feasible to compare the power consumption in other implementations, without the details of the subsystems used in those implementations. The power consumption by the photonic front end for our implementation is less than 160 W. Nevertheless, compared to conventional high-speed electronic ADC systems operating at equivalent bandwidths, the proposed architecture potentially reduces the requirements on the backend electronic sampling rate through photonic time stretching, thereby relaxing the power burden on the electronic digitizer. The proposed neural network for equalization can be implemented in real-time using a GPU or FPGA. For instance, the NVidia RTX PRO 6000 GPU (rated at 125 TOPS [19]) would require a few tens of nanoseconds to run the LSTM + Attention model with a power consumption between 300 - 600 W. Whereas, the Xilinx Virtex UltraScale+ FPGA (rated at 38.3 TOPS [20]) would require a few microseconds to run this model with a power consumption of 15 - 45 W.

Thus, the LSTM-RNN with attention integration allows the PTS front-end to use low-bandwidth electronic ADCs for precisely digitizing high-frequency and ultra-wideband signals, equalizing the distortions, and improving the SNDR and ENOB.

5. Conclusion

We have presented and demonstrated the use of a 1 GSa/s electronic analog-to-digital converter, enabled by a photonic time-stretched frontend integrated with attention-assisted long short-term memory recurrent neural network and a feedforward neural network to detect high-frequency RF signals ranging from 0.6 GHz to 1.4 GHz. The proposed deep neural networks were trained on experimentally collected RF time series data from photonic frontend, effectively equalizes all the system-induced distortions. The long short-term memory recurrent neural network architecture showed an advantage of being able to correct arbitrary lengths of input samples without the need for retraining the model, and the attention mechanism helps model temporal dependencies accurately even for longer signals. The performance of the trained models showed an average frequency and power mismatch of less than 1.9 MHz and less than 0.5 dBm, respectively. The trained architecture with digital reconfigurable bandpass filter integration achieves an increased signal-to-noise and distortion ratio of about 63 dB, equivalent to an effective number of bits of 10.18 bits. This provides a novel method for precisely digitizing ultra-wideband and high-frequency signals with low-bandwidth electronic ADCs.

Funding. Science and Engineering Research Board (IMPRINT-IIC scheme); Department of Telecommunications, India (Advanced Optical Communications Testbed, TTDF/6G/387).

Acknowledgment. The authors would like to thank Sreeraj S J, Joydip Dutta, and Charan Srikanth for the fruitful discussions and valuable inputs that significantly contributed to the development of this work.

Disclosures. The authors declare that there are no conflicts of interest related to this work.

Data availability. The data that support the findings of this study are available from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

References

1. B. Murmann, “ADC performance survey 1997-2023,” (2023). Web: <http://www.stanford.edu/murmann/adcsurvey.html>.
2. Y. Han and B. Jalali, “Photonic time-stretched analog-to-digital converter: fundamental concepts and practical considerations,” *J. Lightwave Technol.* **21**(12), 3085–3103 (2003).
3. Y. Han, O. Boyraz, and B. Jalali, “Ultrawide-band photonic time-stretch a/d converter employing phase diversity,” *IEEE Trans. Microwave Theory Tech.* **53**(4), 1404–1408 (2005).
4. S. Gupta, G. C. Valley, and B. Jalali, “Distortion cancellation in time-stretch analog-to-digital converter,” *J. Lightwave Technol.* **25**(12), 3716–3721 (2007).
5. S. Gupta and B. Jalali, “Time-warp correction and calibration in photonic time-stretch analog-to-digital converter,” *Opt. Lett.* **33**(22), 2674–2676 (2008).
6. B. Xu, W. Lv, J. Ye, *et al.*, “Spurious-free dynamic range improvement in a photonic time-stretched analog-to-digital converter based on third-order predistortion,” *Photonics Res.* **2**(5), 97–101 (2014).
7. S. Yang, J. Wang, H. Chi, *et al.*, “Distortion compensation in continuous-time photonic time-stretched adc based on redundancy detection,” *Appl. Opt.* **60**(6), 1646–1652 (2021).
8. S. Xu and P. R. Prucnal, “Deep-learning-powered photonic analog-to-digital conversion,” *Light: Sci. Appl.* **8**(1), 66 (2019).
9. T. Zhang, S. Hu, L. Zhang, *et al.*, “Multi-channel photonic sampled adc with hybrid deep-learning for distortion recovery,” *Opt. Commun.* **574**, 131192 (2025).
10. M. Singh, J. Dutta, S. S. J., *et al.*, “Device dependent distortion correction in time-stretch photonic analog to digital converters using deep neural networks,” in *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, (2024), pp. 1–3.
11. Q. Liu, L. Long, H. Peng, *et al.*, “Gated spiking neural p systems for time series forecasting,” *IEEE Trans. Neural Netw. Learning Syst.* **34**(9), 6227–6236 (2023).
12. Y. Yang, M. Jin, H. Wen, *et al.*, “A survey on diffusion models for time series and spatio-temporal data,” *ACM Comput. Surv.* **58**(8), 1–39 (2026).
13. R. Cahuantzi, X. Chen, and S. Güttel, “A comparison of lstm and gru networks for learning symbolic sequences,” in *Science and information conference*, (Springer, 2023), pp. 771–785.
14. A. Vaswani, N. Shazeer, N. Parmar, *et al.*, “Attention is all you need,” *Advances in neural information processing systems* **30** (2017).
15. X. Ao, Y. Gong, and A. He, “A review of time series prediction models based on deep learning,” *IEEE Access* (2025).
16. P. Freire, E. Manuylovich, J. E. Prilepsky, *et al.*, “Artificial neural networks for photonic applications—from algorithms to implementation: tutorial,” *Adv. Opt. Photonics* **15**(3), 739–834 (2023).
17. A. Vaswani, N. Shazeer, N. Parmar, *et al.*, “Attention is all you need,” *Advances in Neural Information Processing Systems (NeurIPS)* **30**, 5998–6008 (2017).
18. J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications* (Prentice Hall, 2006), 4th ed.
19. NVIDIA Corporation, “Nvidia rtx pro 6000 blackwell gpu datasheet,” <https://www.nvidia.com/content/dam/en-zz/Solutions/design-visualization/quadro-product-literature/workstation-datasheet-blackwell-rtx-pro6000-x-nvidia-us-3519208-web.pdf> (2025). [Online; accessed Apr. 21, 2026].
20. Xilinx, Inc., “Virtex ultrascale+ vu19p fpga product brief,” <https://docs.amd.com/v/u/en-US/ultrascale-plus-fpga-product-selection-guide> (2020). [Online; accessed Apr. 21, 2026].